

AP[®] Calculus BC Practice Exam

From the 2015 Administration

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A A

CALCULUS BC
SECTION I, Part A
Time—55 minutes
Number of questions—28

A CALCULATOR MAY NOT BE USED ON THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

In this exam:

- (1) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (2) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).

A A

3. $\int x^2 (x^3 + 5)^6 dx =$

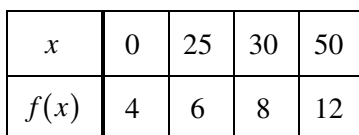
(A) $\frac{1}{3}(x^3 + 5)^6 + C$

(B) $\frac{1}{3}x^3\left(\frac{1}{4}x^4 + 5x\right)^6 + C$

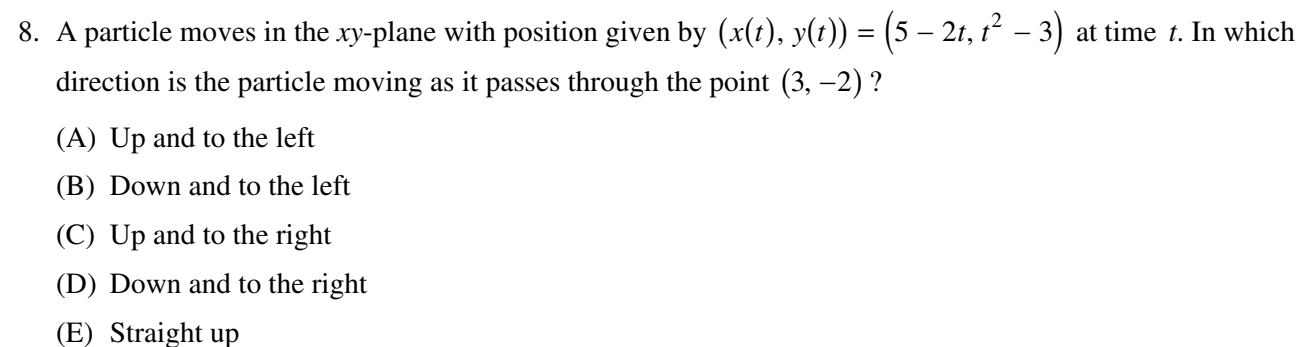
(C) $\frac{1}{7}(x^3 + 5)^7 + C$

(D) $\frac{3}{7}x^2(x^3 + 5)^7 + C$

(E) $\frac{1}{21}(x^3 + 5)^7 + C$



- (E) $\int_1^4 \sqrt{1 + \frac{1}{4}x^2} \, dx$



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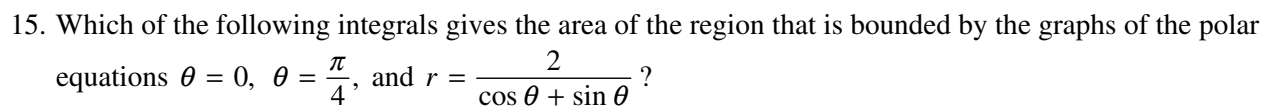
12. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$ is

- (A) ∞ (B) $e - 1$ (C) 1 (D) 0 (E) e^x

13. A population of wolves is modeled by the function P and grows according to the logistic differential equation $\frac{dP}{dt} = 5P\left(1 - \frac{P}{5000}\right)$, where t is the time in years and $P(0) = 1000$. Which of the following statements are true?

- I. $\lim_{t \rightarrow \infty} P(t) = 5000$
- II. $\frac{dP}{dt}$ is positive for $t > 0$.
- III. $\frac{d^2P}{dt^2}$ is positive for $t > 0$.

- (A) I only
(B) II only
(C) I and II only
(D) I and III only
(E) I, II, and III



15. Which of the following integrals gives the area of the region that is bounded by the graphs of the polar equations $\theta = 0$, $\theta = \frac{\pi}{4}$, and $r = \frac{2}{\cos \theta + \sin \theta}$?

$$(E) \int_0^{\pi/4} \frac{2(\cos \theta - \sin \theta)^2}{(\cos \theta + \sin \theta)^4} d\theta$$

20. What is the radius of convergence for the power series $\sum_{n=0}^{\infty} \frac{(x-4)^n}{2 \cdot 3^{n+1}}$?

21. $\int_1^\infty \frac{1}{x^p} dx$ and $\int_0^1 \frac{1}{x^p} dx$ both diverge when $p =$

- (A) 2 (B) 1 (C) $\frac{1}{2}$ (D) 0 (E) -1

22. What are the equations of the horizontal asymptotes of the graph of $y = \frac{2x}{\sqrt{x^2 - 1}}$?

- (A) $y = 0$ only
(B) $y = 1$ only
(C) $y = 2$ only
(D) $y = -2$ and $y = 2$ only
(E) $y = -1$ and $y = 1$ only

23. If $F(x) = \int_4^{x^2} \sqrt{t} \, dt$ for all real numbers $x > 0$, then $F'(x) =$

- (A) $-\frac{1}{2x}$ (B) $\sqrt{x}$ (C) x (D) $2x^2$ (E) $\frac{2x^3 - 16}{3}$

24. Which of the following is the solution to the differential equation $\frac{dy}{dx} = -2xy$ with the initial condition $y(1) = 4$?

(E) $y = e^{-x^2+16}$

28. Let f be a function that has derivatives of all orders for all real numbers, and let $P_3(x)$ be the third-degree Taylor polynomial for f about $x = 0$. The Taylor series for f about $x = 0$ converges at $x = 1$, and

(E) $\frac{3}{4} \cdot \frac{1}{3!}$

-21-

B**B****B****B****B****B****B****B****B****CALCULUS BC****SECTION I, Part B****Time—50 minutes****Number of questions—17**

A GRAPHING CALCULATOR IS REQUIRED FOR SOME QUESTIONS ON
THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

BE SURE YOU ARE USING PAGE 3 OF THE ANSWER SHEET TO RECORD YOUR ANSWERS TO QUESTIONS NUMBERED 76–92.

YOU MAY NOT RETURN TO PAGE 2 OF THE ANSWER SHEET.

In this exam:

- (1) The exact numerical value of the correct answer does not always appear among the choices given. When this happens, select from among the choices the number that best approximates the exact numerical value.
- (2) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (3) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1}x = \arcsin x$).

B**B****B****B****B****B****B****B****B**

76. Let g be a function such that $g(-1) = 0$ and $g(2) = 5$. Which of the following conditions guarantees that there is an x , $-1 < x < 2$, for which $g(x) = 3$?

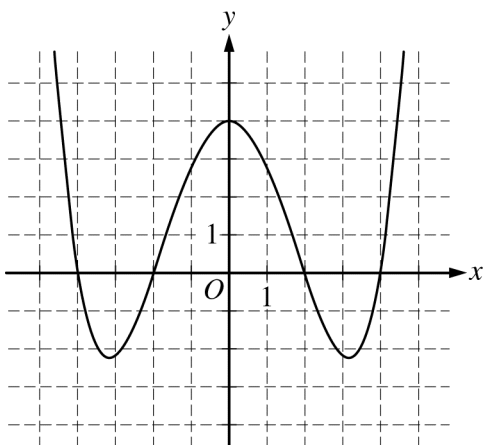
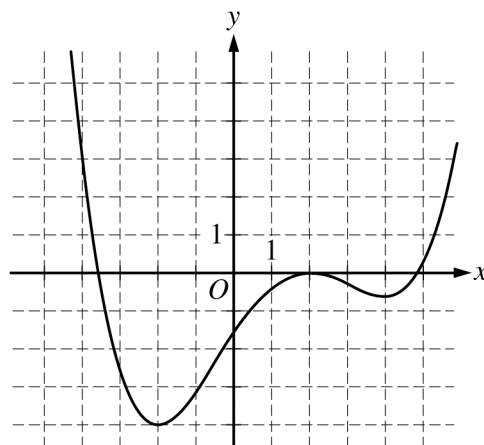
(A) g is defined for all x in $(-1, 2)$.

(B) g is continuous for all x in $[-1, 2]$.

(C) g is increasing on $[-1, 2]$.

(D) There exists an x in $(-1, 2)$ such that $g'(x) = 5$.

(E) $\int_{-1}^2 g(x) dx = 3$

B**B****B****B****B****B****B****B****B**Graph of f Graph of g

77. The graphs of the differentiable functions f and g are shown above. If the function p is defined by $p(x) = f(x)g(x)$, which of the following must be true about p' , the derivative of p ?

- (A) $p'(-2) < 0$
- (B) $p'(-2) = 0$
- (C) $p'(-2) > 0$
- (D) $p'(0) < 0$
- (E) $p'(0) = 0$

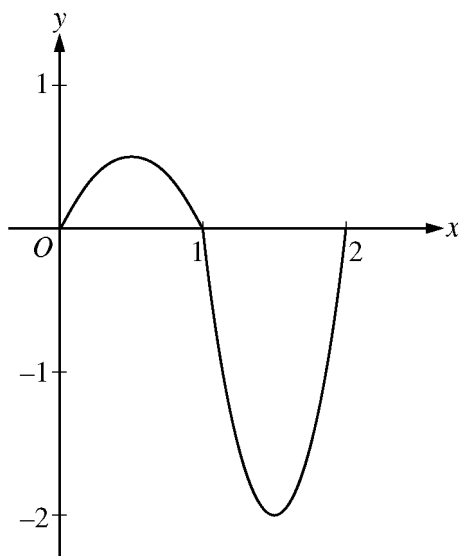
B**B****B****B****B****B****B****B****B**

78. The rate at which motor oil is leaking from an automobile is modeled by the function L defined by $L(t) = 1 + \sin(t^2)$ for time $t \geq 0$. $L(t)$ is measured in liters per hour, and t is measured in hours. How much oil leaks out of the automobile during the first half hour?

- (A) 1.998 liters
- (B) 1.247 liters
- (C) 0.969 liters
- (D) 0.541 liters
- (E) 0.531 liters

-
79. The function f has derivatives of all orders for all real numbers with $f(0) = 3$, $f'(0) = -4$, $f''(0) = 2$, and $f'''(0) = 1$. Let g be the function given by $g(x) = \int_0^x f(t) dt$. What is the third-degree Taylor polynomial for g about $x = 0$?

- (A) $-4x + 2x^2 + \frac{1}{3}x^3$
- (B) $-4x + x^2 + \frac{1}{6}x^3$
- (C) $3x - 2x^2 + \frac{1}{3}x^3$
- (D) $3x - 2x^2 + \frac{2}{3}x^3$
- (E) $3 - 4x + x^2 + \frac{1}{6}x^3$

B**B****B****B****B****B****B****B****B**Graph of f'

80. The figure above shows the graph of f' , the derivative of a function f , for $0 \leq x \leq 2$. What is the value of x at which the absolute minimum of f occurs?

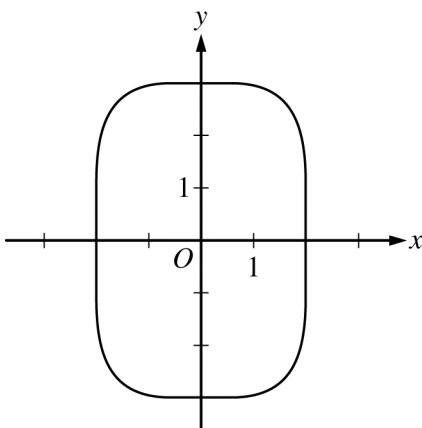
(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) $\frac{3}{2}$

(E) 2

B**B****B****B****B****B****B****B****B**

81. The base of a solid is the region enclosed by the curve $\frac{x^4}{16} + \frac{y^4}{81} = 1$ shown in the figure above. For the solid, each cross section perpendicular to the x -axis is a semicircle. What is the volume of the solid?
- (A) 12.356 (B) 15.732 (C) 22.249 (D) 24.712 (E) 49.425

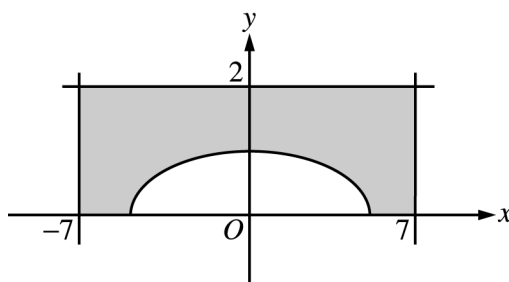
B**B****B****B****B****B****B****B****B**

82. If $f(x) = (x + 2)\sin(\sqrt{x + 2})$, what is the average value of f on the closed interval $[0, 6]$?

- (A) 2.220 (B) 3.348 (C) 4.757 (D) 20.090 (E) 28.541

83. The infinite series $\sum_{k=1}^{\infty} a_k$ has n th partial sum $S_n = \frac{n}{3n+1}$ for $n \geq 1$. What is the sum of the series $\sum_{k=1}^{\infty} a_k$?

- (A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) 1 (D) $\frac{3}{2}$ (E) The series diverges.

B**B****B****B****B****B****B****B****B**

84. The shaded region in the figure above is bounded by the graph of $y = \sqrt{\cos\left(\frac{\pi x}{10}\right)}$ and the lines $x = -7$, $x = 7$, and $y = 2$. What is the area of this region?

- (A) 6.372 (B) 7.628 (C) 20.372 (D) 21.634 (E) 24.923

85. Let $y = f(x)$ define a twice-differentiable function and let $y = t(x)$ be the line tangent to the graph of f at $x = 2$. If $t(x) \geq f(x)$ for all real x , which of the following must be true?

- (A) $f(2) \geq 0$
(B) $f'(2) \geq 0$
(C) $f'(2) \leq 0$
(D) $f''(2) \geq 0$
(E) $f''(2) \leq 0$

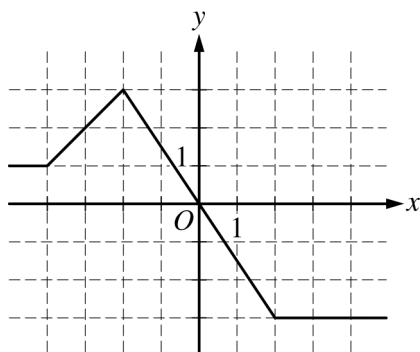
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x	$f(x)$	$f'(x)$	$f''(x)$
0	1	-2	5
1	2	6	-1

86. Let f be a twice-differentiable function with selected values of f and its derivatives shown in the table above.

What is the value of $\int_0^1 x f''(x) \, dx$?

- (A) 6 (B) 5 (C) 3 (D) $-\frac{1}{2}$ (E) -1

B**B****B****B****B****B****B****B****B**Graph of f'

87. The graph of f' , the derivative of the function f , is shown in the figure above. Which of the following statements about f at $x = -2$ is true?

- (A) f is not continuous at $x = -2$.
- (B) f has an absolute maximum at $x = -2$.
- (C) The derivative of f does not exist at $x = -2$.
- (D) The graph of f has a point of inflection at $x = -2$.
- (E) The graph of f has a vertical tangent line at $x = -2$.

88. The first derivative of the function f is given by $f'(x) = \sin(x^2)$. At which of the following values of x does f have a local minimum?

- (A) 2.507 (B) 2.171 (C) 1.772 (D) 1.253 (E) 0

B**B****B****B****B****B****B****B****B**

89. The alternating series test can be used to show convergence of which of the following alternating series?

I. $4 - \frac{1}{9} + 1 - \frac{1}{81} + \frac{1}{4} - \frac{1}{729} + \frac{1}{16} - \cdots + a_n + \cdots$, where $a_n = \begin{cases} \frac{8}{2^n} & \text{if } n \text{ is odd} \\ -\frac{1}{3^n} & \text{if } n \text{ is even} \end{cases}$

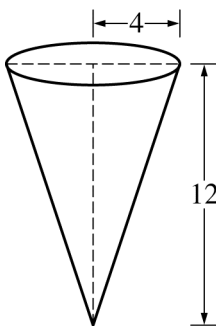
II. $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \cdots + a_n + \cdots$, where $a_n = \frac{(-1)^{n+1}}{n}$

III. $\frac{2}{3} - \frac{3}{5} + \frac{4}{7} - \frac{5}{9} + \frac{6}{11} - \frac{7}{13} + \frac{8}{15} - \cdots + a_n + \cdots$, where $a_n = (-1)^{n+1} \frac{n+1}{2n+1}$

- (A) I only
(B) II only
(C) III only
(D) I and II only
(E) I, II, and III

90. The function f is defined by $f(x) = 3x - 4\cos(2x + 1)$, and its derivative is $f'(x) = 3 + 8\sin(2x + 1)$. What are all values of x that satisfy the conclusion of the Mean Value Theorem applied to f on the interval $[-1, 2]$?

- (A) -0.692 and 1.263 (B) -0.479 and 1.049 (C) 0.285 (D) 0.517 (E) 1.578

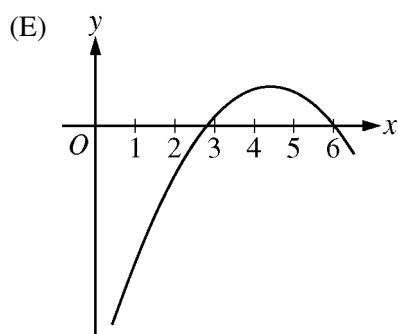
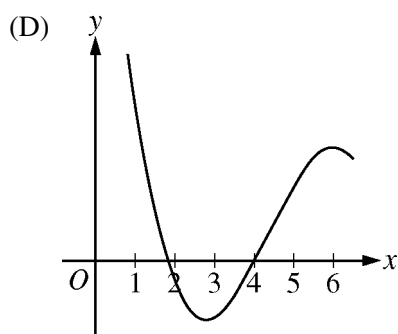
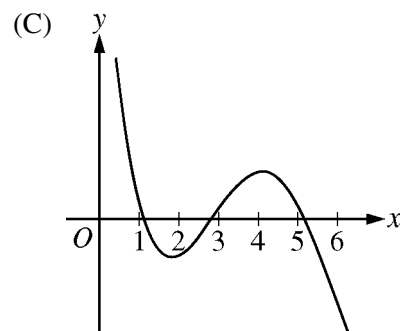
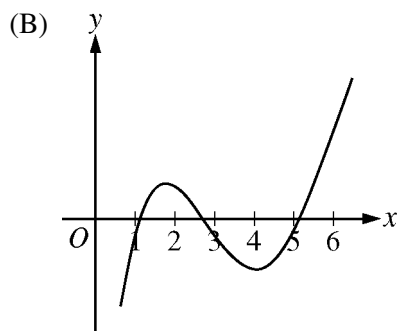
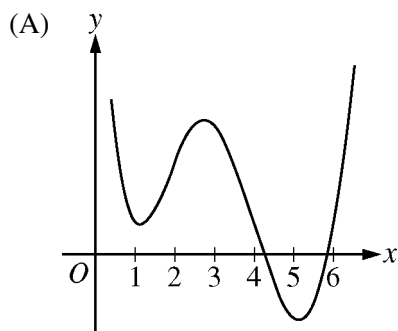
B**B****B****B****B****B****B****B****B**

91. A container has the shape of an open right circular cone, as shown in the figure above. The container has a radius of 4 feet at the top, and its height is 12 feet. If water flows into the container at a constant rate of 6 cubic feet per minute, how fast is the water level rising when the height of the water is 5 feet? (The volume V of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.)
- (A) 0.358 ft/min
(B) 0.688 ft/min
(C) 2.063 ft/min
(D) 8.727 ft/min
(E) 52.360 ft/min

B**B****B****B****B****B****B****B****B**

x	1	2	3	4	5	6
$f(x)$	1	3	4	1	-2	1

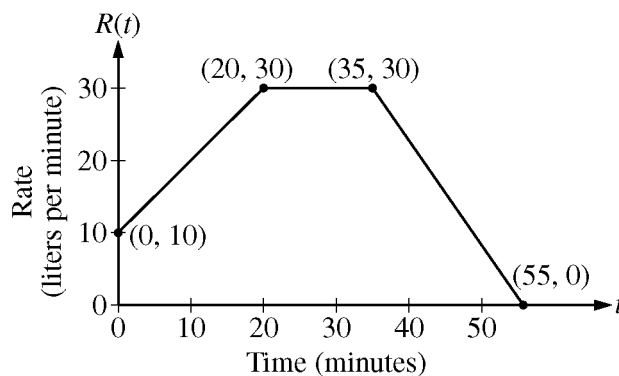
92. The function f is twice differentiable. Selected values of f are given in the table above. Which of the following could be the graph of $f''(x)$, the second derivative of f ?



CALCULUS BC
SECTION II, Part A
Time—30 minutes
Number of problems—2

A graphing calculator is required for these problems.

GO ON TO THE NEXT PAGE.



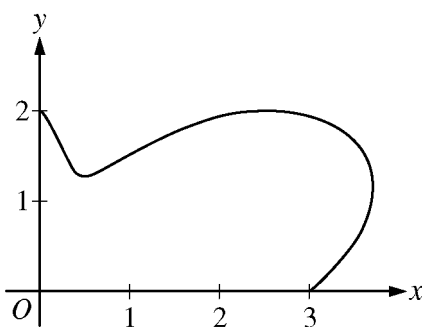
1. At time $t = 0$ minutes, a tank contains 100 liters of water. The piecewise-linear graph above shows the rate $R(t)$, in liters per minute, at which water is pumped into the tank during a 55-minute period.
- (a) Find $R'(45)$. Using appropriate units, explain the meaning of your answer in the context of this problem.

-
- (b) How many liters of water have been pumped into the tank from time $t = 0$ to time $t = 55$ minutes? Show the work that leads to your answer.

1**1****1****1****1****1****1****1****1****1**

- (c) At time $t = 10$ minutes, water begins draining from the tank at a rate modeled by the function D , where $D(t) = 10e^{(\sin t)/10}$ liters per minute. Water continues to drain at this rate until time $t = 55$ minutes. How many liters of water are in the tank at time $t = 55$ minutes?

-
- (d) Using the functions R and D , determine whether the amount of water in the tank is increasing or decreasing at time $t = 45$ minutes. Justify your answer.



2. The figure above shows the graph of the polar equation $r = 2 + \sin(4\theta) + \cos(\theta)$ for $0 \leq \theta \leq \frac{\pi}{2}$. The derivative of r with respect to θ is given by $r'(\theta) = 4\cos(4\theta) - \sin(\theta)$.

(a) Find the area of the region bounded by the graph of r and the lines $\theta = 0$ and $\theta = \frac{\pi}{2}$.

-
- (b) Find the area of the region in the first quadrant that is outside the graph of $r = 2 + \sin(4\theta) + \cos(\theta)$ but inside the graph of the circle of radius 2 centered at the origin.

- (c) Find the value of θ in the interval $0 \leq \theta \leq \frac{\pi}{2}$ that corresponds to the point on the curve $r = 2 + \sin(4\theta) + \cos(\theta)$ with greatest distance from the origin. Justify your answer.

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CALCULUS BC
SECTION II, Part B
Time—60 minutes
Number of problems—4

No calculator is allowed for these problems.

DO NOT BREAK THE SEALS UNTIL YOU ARE TOLD TO DO SO.

GO ON TO THE NEXT PAGE.

NO CALCULATOR ALLOWED

t (seconds)	0	3	5	8	12
$k(t)$ (feet per second)	0	5	10	20	24

3. Kathleen skates on a straight track. She starts from rest at the starting line at time $t = 0$. For $0 < t \leq 12$ seconds, Kathleen's velocity k , measured in feet per second, is differentiable and increasing. Values of $k(t)$ at various times t are given in the table above.
- (a) Use the data in the table to estimate Kathleen's acceleration at time $t = 4$ seconds. Show the computations that lead to your answer. Indicate units of measure.

-
- (b) Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate $\int_0^{12} k(t) dt$. Indicate units of measure. Is this approximation an overestimate or an underestimate for the value of $\int_0^{12} k(t) dt$? Explain your reasoning.

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NO CALCULATOR ALLOWED

- (c) Nathan skates on the same track, starting 5 feet ahead of Kathleen at time $t = 0$. Nathan's velocity, in feet per second, is given by $n(t) = \frac{150}{t+3} - 50e^{-t}$. Write, but do not evaluate, an expression involving an integral that gives Nathan's distance from the starting line at time $t = 12$ seconds.

-
- (d) Write an expression for Nathan's acceleration in terms of t .

4**4****4****4****4****4****4****4****4****4****NO CALCULATOR ALLOWED**

4. In a national park, the population of mountain lions grows over time. At time $t = 0$, where t is measured in years, the population is found to be 20 mountain lions.

(a) One zoologist suggests a population model P that satisfies the differential equation $\frac{dP}{dt} = \frac{1}{4}(220 - P)$.

Use separation of variables to solve this differential equation for P with the initial condition $P(0) = 20$.

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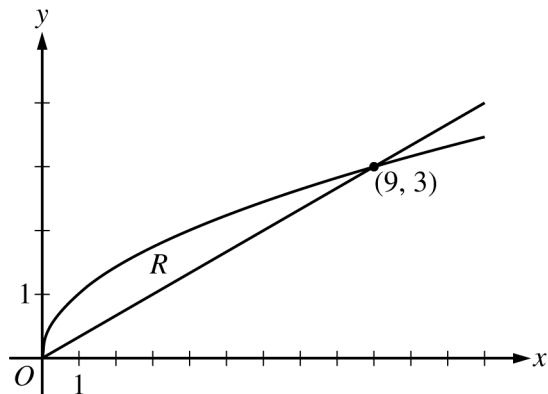
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NO CALCULATOR ALLOWED

- (b) A second zoologist suggests a population model Q that satisfies $\frac{dQ}{dt} = \frac{1}{500}Q(220 - Q)$. Find the value of $\frac{dQ}{dt}$ at the time when Q grows most rapidly.

-
- (c) For the population model Q introduced in part (b), use Euler's method, starting at $t = 0$ with two steps of equal size, to approximate $Q(10)$. Show the computations that lead to your answer.

NO CALCULATOR ALLOWED



5. Let R be the region in the first quadrant enclosed by the graphs of $g(x) = \sqrt{x}$ and $h(x) = \frac{x}{3}$, as shown in the figure above.
- (a) Find the area of region R .

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NO CALCULATOR ALLOWED

- (b) Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid generated when R is revolved about the horizontal line $y = 4$.

-
- (c) Find the maximum vertical distance between the graph of g and the graph of h between $x = 0$ and $x = 16$. Justify your answer.

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NO CALCULATOR ALLOWED

6. The Maclaurin series for a function f is given by $\frac{x}{3} + \frac{x^2}{4} + \frac{x^3}{5} + \cdots + \frac{x^n}{n+2} + \cdots$.

(a) Use the ratio test to find the interval of convergence of the Maclaurin series for f .

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NO CALCULATOR ALLOWED

- (b) Let g be the function given by $g(x) = f(-2x)$. Find the first three terms and the general term of the Maclaurin series for g .

-
- (c) The first two terms of the Maclaurin series for f are used to approximate $f(0.1)$. Given that $|f'''(x)| \leq 2$ for $0 \leq x \leq 0.1$, use the Lagrange error bound to show that this approximation differs from $f(0.1)$ by at most $\frac{1}{3000}$.

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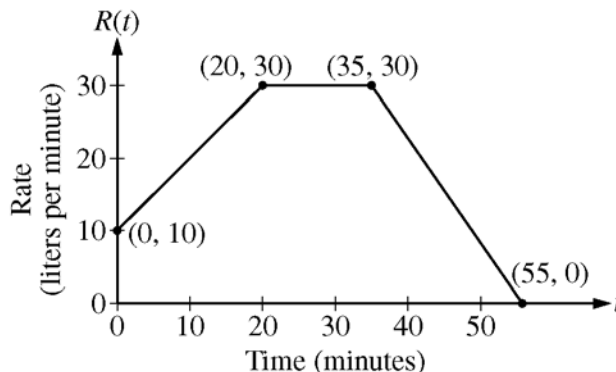
Answer Key for AP Calculus BC Practice Exam, Section I

Question 1: E	Question 24: D
Question 2: B	Question 25: B
Question 3: E	Question 26: A
Question 4: A	Question 27: A
Question 5: D	Question 28: B
Question 6: B	Question 76: B
Question 7: D	Question 77: A
Question 8: A	Question 78: D
Question 9: C	Question 79: C
Question 10: E	Question 80: E
Question 11: B	Question 81: E
Question 12: C	Question 82: B
Question 13: C	Question 83: A
Question 14: D	Question 84: C
Question 15: C	Question 85: E
Question 16: B	Question 86: B
Question 17: B	Question 87: D
Question 18: B	Question 88: A
Question 19: C	Question 89: B
Question 20: C	Question 90: B
Question 21: B	Question 91: B
Question 22: D	Question 92: D
Question 23: D	

AP[®] CALCULUS BC
2015 SCORING GUIDELINES

Question 1

At time $t = 0$ minutes, a tank contains 100 liters of water. The piecewise-linear graph above shows the rate $R(t)$, in liters per minute, at which water is pumped into the tank during a 55-minute period.



- (a) Find $R'(45)$. Using appropriate units, explain the meaning of your answer in the context of this problem.
- (b) How many liters of water have been pumped into the tank from time $t = 0$ to time $t = 55$ minutes? Show the work that leads to your answer.
- (c) At time $t = 10$ minutes, water begins draining from the tank at a rate modeled by the function D , where $D(t) = 10e^{(\sin t)/10}$ liters per minute. Water continues to drain at this rate until time $t = 55$ minutes. How many liters of water are in the tank at time $t = 55$ minutes?
- (d) Using the functions R and D , determine whether the amount of water in the tank is increasing or decreasing at time $t = 45$ minutes. Justify your answer.

(a) $R'(45) = \frac{30 - 0}{35 - 55} = -\frac{3}{2}$

The rate at which water is being pumped into the tank is decreasing at $\frac{3}{2}$ liters/min² at $t = 45$ minutes.

(b)
$$\int_0^{55} R(t) dt = 20 \cdot \frac{10 + 30}{2} + 15 \cdot 30 + \frac{1}{2} \cdot 20 \cdot 30$$

$$= 400 + 450 + 300 = 1150$$

(c)
$$\text{Amt} = 100 + 1150 - \int_{10}^{55} 10e^{(\sin t)/10} dt$$

$$= 1250 - 450.275371 = 799.725 \text{ (or } 799.724)$$

(d) $R(45) = 15$
 $D(45) = 10.88815$

At time $t = 45$ minutes, the rate of water pumped into the tank is greater than the rate of water draining from the tank. Therefore, the amount of water in the tank is increasing at time $t = 45$ minutes.

2 : $\begin{cases} 1 : R'(45) \\ 1 : \text{explanation} \end{cases}$

2 : $\begin{cases} 1 : \text{sum of areas} \\ 1 : \text{answer} \end{cases}$

3 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{expression for water in the tank} \\ 1 : \text{answer} \end{cases}$

2 : answer with justification

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Question 2

The figure above shows the graph of the polar equation

$r = 2 + \sin(4\theta) + \cos(\theta)$ for $0 \leq \theta \leq \frac{\pi}{2}$. The derivative of r with

respect to θ is given by $r'(\theta) = 4\cos(4\theta) - \sin(\theta)$.

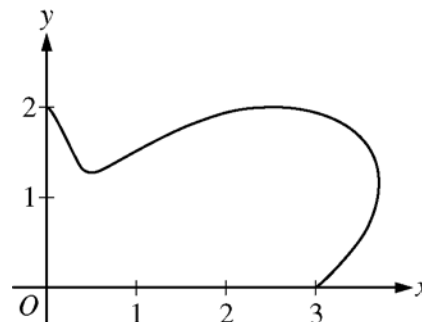
- (a) Find the area of the region bounded by the graph of r and the lines

$$\theta = 0 \text{ and } \theta = \frac{\pi}{2}.$$

- (b) Find the area of the region in the first quadrant that is outside the graph of $r = 2 + \sin(4\theta) + \cos(\theta)$ but inside the graph of the circle of radius 2 centered at the origin.

- (c) Find the value of θ in the interval $0 \leq \theta \leq \frac{\pi}{2}$ that corresponds to the point on the curve

$r = 2 + \sin(4\theta) + \cos(\theta)$ with greatest distance from the origin. Justify your answer.



(a)
$$\text{Area} = \frac{1}{2} \int_0^{\pi/2} (2 + \sin(4\theta) + \cos(\theta))^2 d\theta$$

$$= 6.194 \text{ (or } 6.193)$$

2 : $\begin{cases} 1 : \text{integrand} \\ 1 : \text{answer} \end{cases}$

(b) $2 + \sin(4\theta) + \cos(\theta) = 2 \Rightarrow \theta = 0.942478$
 Let $c = 0.942478$

$$\text{Area} = \frac{1}{2} \int_c^{\pi/2} [2^2 - (2 + \sin(4\theta) + \cos(\theta))^2] d\theta$$

$$= 0.456$$

4 : $\begin{cases} 1 : \text{value of } \theta \\ 2 : \text{integrand} \\ 1 : \text{answer} \end{cases}$

(c) $r'(\theta) = 4\cos(4\theta) - \sin(\theta) = 0$
 $\Rightarrow \theta = 0.370064, 1.237726$

3 : $\begin{cases} 1 : \text{identifies } \theta = 0.370 \text{ as a candidate} \\ 2 : \text{answer with justification} \end{cases}$

θ	$r(\theta)$
0	3
0.370064	3.928208
1.237726	1.355256
$\pi/2$	2

$\theta = 0.370$ corresponds to the point on the curve with the greatest distance from the origin.

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Question 3

t (seconds)	0	3	5	8	12
$k(t)$ (feet per second)	0	5	10	20	24

Kathleen skates on a straight track. She starts from rest at the starting line at time $t = 0$. For $0 < t \leq 12$ seconds, Kathleen's velocity k , measured in feet per second, is differentiable and increasing. Values of $k(t)$ at various times t are given in the table above.

- (a) Use the data in the table to estimate Kathleen's acceleration at time $t = 4$ seconds. Show the computations that lead to your answer. Indicate units of measure.
- (b) Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate $\int_0^{12} k(t) dt$. Indicate units of measure. Is this approximation an overestimate or an underestimate for the value of $\int_0^{12} k(t) dt$? Explain your reasoning.
- (c) Nathan skates on the same track, starting 5 feet ahead of Kathleen at time $t = 0$. Nathan's velocity, in feet per second, is given by $n(t) = \frac{150}{t+3} - 50e^{-t}$. Write, but do not evaluate, an expression involving an integral that gives Nathan's distance from the starting line at time $t = 12$ seconds.
- (d) Write an expression for Nathan's acceleration in terms of t .

(a) $a(4) \approx \frac{10 - 5}{5 - 3} = \frac{5}{2} \text{ ft/sec}^2$

2 : $\begin{cases} 1 : \text{estimate} \\ 1 : \text{units} \end{cases}$

(b) $\int_0^{12} k(t) dt \approx (5)(3) + (10)(2) + (20)(3) + (24)(4) = 191 \text{ feet}$

This approximation is an overestimate since a right Riemann sum is used and the function k is increasing.

3 : $\begin{cases} 1 : \text{right Riemann sum} \\ 1 : \text{approximation with units} \\ 1 : \text{overestimate with reason} \end{cases}$

(c) $s(12) = 5 + \int_0^{12} n(t) dt$

2 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

(d) $n'(t) = (150)(-1)(t+3)^{-2} - 50e^{-t}(-1)$
 $= -\frac{150}{(t+3)^2} + 50e^{-t}$

2 : $n'(t)$

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Question 4

In a national park, the population of mountain lions grows over time. At time $t = 0$, where t is measured in years, the population is found to be 20 mountain lions.

- (a) One zoologist suggests a population model P that satisfies the differential equation $\frac{dP}{dt} = \frac{1}{4}(220 - P)$.
 Use separation of variables to solve this differential equation for P with the initial condition $P(0) = 20$.
- (b) A second zoologist suggests a population model Q that satisfies $\frac{dQ}{dt} = \frac{1}{500}Q(220 - Q)$. Find the value of $\frac{dQ}{dt}$ at the time when Q grows most rapidly.
- (c) For the population model Q introduced in part (b), use Euler's method, starting at $t = 0$ with two steps of equal size, to approximate $Q(10)$. Show the computations that lead to your answer.

(a) $\frac{dP}{dt} = \frac{1}{4}(220 - P)$

$$\int \frac{dP}{220 - P} = \int \frac{1}{4} dt$$

$$-\ln|220 - P| = \frac{1}{4}t + C$$

Because $P(0) = 20$, $P < 220$, so $|220 - P| = 220 - P$.

$$-\ln(220 - 20) = \frac{1}{4}(0) + C \Rightarrow C = -\ln 200$$

$$220 - P = 200e^{-t/4}$$

$$P = 220 - 200e^{-t/4}, t \geq 0$$

$$5 : \begin{cases} 1 : \text{separation of variables} \\ 1 : \text{antiderivatives} \\ 1 : \text{constant of integration} \\ 1 : \text{uses initial condition} \\ 1 : \text{solves for } P \end{cases}$$

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

- (b) Q satisfies a logistic differential equation with carrying capacity 220. Q grows most rapidly when $Q = \frac{220}{2} = 110$.

$$\left. \frac{dQ}{dt} \right|_{Q=110} = \frac{110^2}{500} = \frac{121}{5}$$

$$2 : \begin{cases} 1 : Q = 110 \\ 1 : \text{answer} \end{cases}$$

(c) $Q(0) = 20$

$$Q'(0) = \frac{1}{500}(20)(200) = 8$$

$$Q(5) \approx 20 + (8)(5) = 60$$

$$Q'(5) \approx \frac{1}{500}(60)(220 - 60) = \frac{96}{5}$$

$$Q(10) \approx 60 + \left(\frac{96}{5}\right)(5) = 156$$

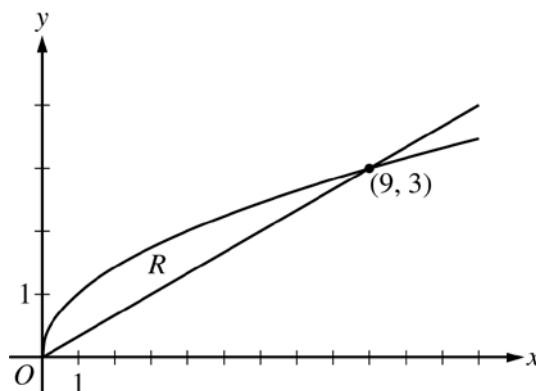
$$2 : \begin{cases} 1 : \text{Euler's method with two steps} \\ 1 : \text{answer} \end{cases}$$

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Question 5

Let R be the region in the first quadrant enclosed by the graphs of $g(x) = \sqrt{x}$ and $h(x) = \frac{x}{3}$, as shown in the figure above.

- Find the area of region R .
- Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid generated when R is revolved about the horizontal line $y = 4$.
- Find the maximum vertical distance between the graph of g and the graph of h between $x = 0$ and $x = 16$. Justify your answer.



$$\begin{aligned} \text{(a) Area} &= \int_0^9 \left(\sqrt{x} - \frac{x}{3} \right) dx = \left[\frac{2}{3} x^{3/2} - \frac{1}{6} x^2 \right]_0^9 \\ &= \frac{2}{3} \cdot 27 - \frac{1}{6} \cdot 81 = \frac{9}{2} \end{aligned}$$

$$3 : \begin{cases} 1 : \text{integrand} \\ 1 : \text{antiderivative} \\ 1 : \text{answer} \end{cases}$$

$$\text{(b) Volume} = \pi \int_0^9 \left[\left(4 - \frac{x}{3} \right)^2 - \left(4 - \sqrt{x} \right)^2 \right] dx$$

$$3 : \begin{cases} 2 : \text{integrand} \\ 1 : \text{limits and constant} \end{cases}$$

(c) Consider the function $D(x) = \sqrt{x} - \frac{x}{3}$.

$$D'(x) = \frac{1}{2} x^{-1/2} - \frac{1}{3} = \frac{1}{2\sqrt{x}} - \frac{1}{3}$$

$$D'(x) = 0 \Rightarrow x = \frac{9}{4}$$

$$3 : \begin{cases} 1 : \text{sets } D'(x) = 0 \\ 1 : \text{identifies } x = \frac{9}{4} \text{ as a candidate} \\ 1 : \text{answer and justification} \end{cases}$$

x	$D(x)$	Distance between graphs
0	0	0
$\frac{9}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
16	$-\frac{4}{3}$	$\frac{4}{3}$

The maximum vertical distance between the graph of g and the graph of h between $x = 0$ and $x = 16$ is $\frac{4}{3}$.

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Question 6

The Maclaurin series for a function f is given by $\frac{x}{3} + \frac{x^2}{4} + \frac{x^3}{5} + \cdots + \frac{x^n}{n+2} + \cdots$.

- (a) Use the ratio test to find the interval of convergence of the Maclaurin series for f .
- (b) Let g be the function given by $g(x) = f(-2x)$. Find the first three terms and the general term of the Maclaurin series for g .
- (c) The first two terms of the Maclaurin series for f are used to approximate $f(0.1)$. Given that $|f'''(x)| \leq 2$ for $0 \leq x \leq 0.1$, use the Lagrange error bound to show that this approximation differs from $f(0.1)$ by at most $\frac{1}{3000}$.

- (a) Let a_n be the n th term of the Maclaurin series.

$$\frac{a_{n+1}}{a_n} = \frac{x^{n+1}}{n+3} \cdot \frac{n+2}{x^n} = \left(\frac{n+2}{n+3}\right) \cdot x$$

$$\lim_{n \rightarrow \infty} \left| \left(\frac{n+2}{n+3}\right) \cdot x \right| = |x|$$

$$|x| < 1 \Rightarrow -1 < x < 1$$

The series converges when $-1 < x < 1$.

When $x = -1$, the series is $-\frac{1}{3} + \frac{1}{4} - \frac{1}{5} + \frac{1}{6} - \cdots$.

This series converges by the alternating series test.

When $x = 1$, the series is $\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \cdots$.

This series diverges since it is a harmonic series.

Therefore, the interval of convergence is $-1 \leq x < 1$.

- (b) The first three terms are

$$\frac{-2x}{3} + \frac{(-2x)^2}{4} + \frac{(-2x)^3}{5} = -\frac{2}{3}x + x^2 - \frac{8}{5}x^3.$$

The general term is $\frac{(-2x)^n}{n+2}$.

- (c) Error $\leq \frac{\max_{0 \leq x \leq 0.1} |f'''(x)|}{3!} \cdot \left(\frac{1}{10}\right)^3 \leq \frac{2}{6} \cdot \frac{1}{1000} = \frac{1}{3000}$

5 : { 1 : sets up ratio
1 : computes limit of ratio
1 : identifies interior of interval of convergence
1 : considers both endpoints
1 : analysis and interval of convergence

2 : { 1 : first three terms
1 : general term

2 : { 1 : form of the error bound
1 : analysis